

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{18} g_2^4 (g_1^2 + g_2^2) LF_{3,0}[m_2] + \frac{1}{12} g_2^4 (g_1^2 + g_2^2) LF_{4,-1}[m_2] - \frac{4}{45} g_2^4 (g_1^2 + g_2^2) LF_{5,-2}[m_2] + \right. \\
& \frac{1}{216} \sum_p g_1^6 LF_{3,0}[m_d^p] + \frac{1}{24} \sum_p g_1^6 LF_{3,0}[m_e^p] - \frac{1}{288} \sum_p (3 g_1^6 + 19 g_1^2 g_2^4 + 10 g_2^6) LF_{3,0}[m_l^p] + \\
& \frac{1}{96} \sum_p g_2^4 (g_1^2 + g_2^2) LF_{4,-1}[m_l^p] + \frac{1}{45} \sum_p g_2^4 (g_1^2 + g_2^2) LF_{5,-2}[m_l^p] + \\
& \frac{1}{864} (18 \overline{y_u^{pr}} y_u^{pr} (g_1^4 + 30 g_1^2 g_2^2 + 45 g_2^4) + \sum_p (g_1^6 - 63 g_1^2 g_2^4 - 90 g_2^6)) LF_{3,0}[m_q^p] + \\
& \frac{1}{32} g_2^2 (g_1^2 + g_2^2) (-24 \overline{y_u^{pr}} y_u^{pr} + \sum_p g_2^2) LF_{4,-1}[m_q^p] + \frac{1}{15} \sum_p g_2^4 (g_1^2 + g_2^2) LF_{5,-2}[m_q^p] - \\
& \frac{1}{27} \sum_p g_1^6 LF_{3,0}[m_u^p] + \frac{1}{3} g_1^4 \overline{y_u^{pr}} y_u^{pr} LF_{3,0}[m_u^r] + \frac{1}{36} g_2^4 (g_1^2 + g_2^2) LF_{3,0}[\tilde{\mu}] + \\
& \frac{1}{24} g_2^4 (g_1^2 + g_2^2) LF_{4,-1}[\tilde{\mu}] - \frac{2}{45} g_2^4 (g_1^2 + g_2^2) LF_{5,-2}[\tilde{\mu}] - \frac{1}{12} g_1^6 LF_{3,3,-3}[m_1, \tilde{\mu}] - \\
& \frac{1}{4} g_1^6 m_1^2 LF_{3,3,-2}[m_1, \tilde{\mu}] - \frac{1}{3} g_2^4 (g_1^2 + g_2^2) LF_{2,1,0}[m_2, \tilde{\mu}] - \frac{17}{24} g_2^4 (g_1^2 + g_2^2) LF_{2,2,-1}[m_2, \tilde{\mu}] + \\
& \frac{1}{6} g_2^4 (g_1^2 + g_2^2) LF_{3,1,-1}[m_2, \tilde{\mu}] + \frac{7}{6} g_2^4 (g_1^2 + g_2^2) LF_{3,2,-2}[m_2, \tilde{\mu}] - \\
& \frac{3}{4} g_2^4 m_2^2 (g_1^2 + g_2^2) LF_{3,2,-1}[m_2, \tilde{\mu}] - \frac{1}{4} g_2^4 (2 g_1^2 + 5 g_2^2) LF_{3,3,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{4} g_2^4 m_2^2 (2 g_1^2 + g_2^2) LF_{3,3,-2}[m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 (g_1^2 + g_2^2) LF_{4,2,-3}[m_2, \tilde{\mu}] + \\
& \frac{1}{2} g_2^4 m_2^2 (g_1^2 + g_2^2) LF_{4,2,-2}[m_2, \tilde{\mu}] - \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} LF_{2,2,0}[m_d^r, m_q^p] + \\
& \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} LF_{3,1,0}[m_d^r, m_q^p] - \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} LF_{2,2,0}[m_e^r, m_l^p] + \\
& \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} LF_{3,1,0}[m_e^r, m_l^p] + \frac{1}{4} g_1^4 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} LF_{3,1,0}[m_l^p, m_e^r] + \\
& \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} (g_1^2 + g_2^2) LF_{3,2,-1}[m_l^p, m_e^r] - \frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} (g_1^2 + g_2^2) \\
& LF_{4,1,-1}[m_l^p, m_e^r] + \frac{1}{24} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) LF_{5,1,-2}[m_l^p, m_e^r] + \\
& \frac{1}{12} g_1^4 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} LF_{3,1,0}[m_q^p, m_d^r] + \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) LF_{3,2,-1}[m_q^p, m_d^r] - \\
& \frac{3}{8} g_1^2 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} (g_1^2 + g_2^2) LF_{4,1,-1}[m_q^p, m_d^r] + \frac{1}{8} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} (3 g_1^2 - g_2^2) \\
& (g_1^2 + g_2^2) LF_{5,1,-2}[m_q^p, m_d^r] - \frac{1}{2} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + 3 g_2^2) LF_{2,1,0}[m_q^p, m_q^s] + \\
& \frac{3}{4} \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} (g_1^2 + g_2^2) LF_{3,1,-1}[m_q^p, m_q^s] + \frac{1}{48} \overline{a_u^{pr}} a_u^{pr} (-7 g_1^4 + 30 g_1^2 g_2^2 + 21 g_2^4) \\
& LF_{2,2,0}[m_q^p, m_u^r] + \frac{1}{48} \overline{a_u^{pr}} a_u^{pr} (g_1^4 + 30 g_1^2 g_2^2 + 45 g_2^4) LF_{3,1,0}[m_q^p, m_u^r] - \\
& \frac{1}{2} g_2^2 \overline{a_u^{pr}} a_u^{pr} (g_1^2 + g_2^2) LF_{3,2,-1}[m_q^p, m_u^r] - \frac{3}{4} g_2^2 \overline{a_u^{pr}} a_u^{pr} (g_1^2 + g_2^2) LF_{4,1,-1}[m_q^p, m_u^r] + \\
& \frac{1}{48} \overline{a_u^{pr}} a_u^{pr} (49 g_1^4 + 30 g_1^2 g_2^2 - 3 g_2^4) LF_{3,1,0}[m_u^r, m_q^p] + \\
& \frac{1}{16} \overline{a_u^{pr}} a_u^{pr} (g_1^2 - 7 g_2^2) (g_1^2 + g_2^2) LF_{3,2,-1}[m_u^r, m_q^p] - \\
& \frac{3}{16} \overline{a_u^{pr}} a_u^{pr} (7 g_1^2 - g_2^2) (g_1^2 + g_2^2) LF_{4,1,-1}[m_u^r, m_q^p] + \\
& \frac{1}{8} \overline{a_u^{pr}} a_u^{pr} (3 g_1^2 - g_2^2) (g_1^2 + g_2^2) LF_{5,1,-2}[m_u^r, m_q^p] - \\
& g_1^2 \overline{y_u^{pr}} \overline{y_u^{st}} y_u^{pt} y_u^{sr} LF_{2,1,0}[m_u^r, m_u^t] - \frac{1}{8} g_1^2 LF_{3,1,-1}[\tilde{\mu}, m_1] (g_1^2 + g_2^2)^2 + \\
& \frac{3}{16} g_1^4 (g_1^2 + g_2^2) LF_{3,2,-2}[\tilde{\mu}, m_1] + \frac{3}{16} g_1^4 m_1^2 (g_1^2 + g_2^2) LF_{3,2,-1}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} g_1^2 (g_1^2 + g_2^2) (6 g_1^2 + g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 (g_1^2 + g_2^2) LF_{4,2,-3}[\tilde{\mu}, m_1] - \\
& \frac{1}{8} g_1^4 m_1^2 (g_1^2 + g_2^2) LF_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{24} (-3 g_1^6 - 2 g_1^4 g_2^2 + g_1^2 g_2^4) LF_{5,1,-3}[\tilde{\mu}, m_1] + \\
& \frac{1}{24} (-9 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + 7 g_2^6) LF_{3,1,-1}[\tilde{\mu}, m_2] + \\
& \frac{85}{48} g_2^4 (g_1^2 + g_2^2) LF_{3,2,-2}[\tilde{\mu}, m_2] + \frac{3}{16} g_2^4 m_2^2 (g_1^2 + g_2^2) LF_{3,2,-1}[\tilde{\mu}, m_2] + \\
& \frac{3}{8} (2 g_1^4 g_2^2 + g_1^2 g_2^4 - g_2^6) LF_{4,1,-2}[\tilde{\mu}, m_2] - \frac{7}{8} g_2^4 (g_1^2 + g_2^2) LF_{4,2,-3}[\tilde{\mu}, m_2] - \\
& \frac{1}{8} g_2^4 m_2^2 (g_1^2 + g_2^2) LF_{4,2,-2}[\tilde{\mu}, m_2] + \frac{1}{8} (-3 g_1^4 g_2^2 - 2 g_1^2 g_2^4 + g_2^6) LF_{5,1,-3}[\tilde{\mu}, m_2] + \\
& g_1^2 g_2^2 (g_1^2 + g_2^2) LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_1] + \frac{3}{4} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
& \frac{1}{2} g_1^2 g_2^2 (g_1^2 + g_2^2) LF_{3,2,1,-3}[m_2, \tilde{\mu}, m_1] - \frac{1}{2} m_1 m_2 g_1^2 g_2^2 (g_1^2 + g_2^2) LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{sr} \overline{y_u^{st}} y_u^{pt} (g_1^2 + g_2^2) LF_{2,2,1,-1}[m_q^p, m_q^s, m_d^r] + \\
& \overline{y_u^{pr}} \overline{y_u^{st}} \overline{y_u^{uv}} y_u^{pt} y_u^{sv} y_u^{ur} LF_{1,1,1,0}[m_q^p, m_q^s, m_q^u] - \frac{1}{2} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} a_u^{sr} (g_1^2 + 3 g_2^2) \\
& LF_{2,1,1,0}[m_q^p, m_q^s, m_u^r] + \frac{3}{4} \overline{y_u^{st}} y_u^{pt} \overline{a_u^{pr}} a_u^{sr} (g_1^2 + g_2^2) LF_{3,1,1,-1}[m_q^p, m_q^s, m_u^r] + \\
& \frac{3}{8} \overline{y_u^{st}} y_u^{pt} \overline{a_u$$